

Phase 8

Project Demonstration Phase

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The project demonstration video provides a complete walkthrough of the FitTrack AI backend application, covering the modular codebase structure, live server execution, and end-to-end REST API testing.

Project File & Architecture Walkthrough:

Explains the Model-View-Controller (MVC) folder structure inside Visual Studio Code. This includes the database configuration (`config/db.js`), Mongoose data models (`models/User.js` and `models/Workout.js`), Express route definitions (`routes/authRoutes.js`, `routes/workoutRoutes.js`, and `routes/aiRoutes.js`), business logic controllers (`controllers/authController.js`, `controllers/workoutController.js`, and `controllers/aiController.js`), JWT authentication and error-handling middleware (`middleware/`), and the Google Gemini service layer (`services/geminiService.js`).

Backend & Database Execution:

Demonstrates configuring the `.env` file with the server port, MongoDB URI, JWT secret, and Google Gemini API key. It then shows starting the Node.js and Express.js server via the terminal (`npm start`) and verifying a successful connection to the local MongoDB database.

Live API Endpoint Testing:

Showcases real-time HTTP request and JSON response verification using Thunder Client and Postman across three core modules:

- **Authentication APIs:** Testing user registration (`POST /api/auth/register`), user login (`POST /api/auth/login`) to generate a stateless JWT token, and protected profile retrieval (`GET /api/auth/profile`) using Bearer token authorization.
- **Workout Management APIs:** Executing protected CRUD operations, including adding a new workout (`POST /api/workouts/`), retrieving all user workouts (`GET /api/workouts`), fetching a specific workout by ID (`GET /api/workouts/:id`), updating workout details (`PUT /api/workouts/:id`), and deleting a workout record (`DELETE /api/workouts/:id`).
- **Google Gemini AI APIs:** Sending user fitness parameters to generate tailored workout plans (`POST /api/ai/workout-recommendation`) and submitting workout metrics to receive AI-analyzed performance summaries (`POST /api/ai/fitness-insights`).

Demo: [Demo Video Link](#)